



Module 5: Practical Circuits

Digital Circuits · Amplifiers · Filters · SDR · Op-Amps · Oscillators
2024-2028 Amateur Extra Question Pool



The Heaviest Module

- 8 exam questions — E7A through E7H — 16% of your Extra exam
- Tied with E9 as the heaviest subelement in the pool
- This is the hardest module — the content spans RF circuit design
- Hour 1:
 - → E7A: Digital circuits — flip-flops, counters, logic gates
 - → E7B: Amplifier classes (A, AB, B, C, D)
 - → E7C: Filter responses and matching networks
- Hour 2:
 - → E7D-E: Power supplies, modulation/demodulation
 - → E7F: Software Defined Radio and DSP
 - → E7G: Operational amplifiers and gain formulas
 - → E7H: Oscillators — Colpitts, Hartley, PLL, DDS



Part 1: Digital Circuits

Subelement E7A — 1 exam question



Flip-Flops, Counters, Multivibrators

- Flip-flop: bistable circuit — stays in state 0 or 1 until a clock edge triggers it
- → Each flip-flop divides frequency by 2
- → 4 flip-flops in series: divide by $2^4 = 16$
- Decade counter: produces one output pulse per 10 input pulses ($\div 10$)
- Astable multivibrator:
 - → Continuously oscillates — no stable state
 - → Generates its own square wave — no external clock needed
- Monostable multivibrator (one-shot):
 - → Has ONE stable state
 - → Trigger pulse sends it to the other state for a fixed time, then returns
- Bistable (flip-flop): requires external trigger to change state



Logic Gate Truth Tables

Gate Behavior

- NAND: output 0 ONLY when all inputs are 1
- NOR: output 1 ONLY when all inputs are 0
- OR: output 1 if ANY input is 1
- AND: output 1 ONLY when all inputs are 1
- XOR: output 1 when inputs DIFFER
- XNOR: output 0 when inputs differ
- NOT: output is opposite of input
- NAND = universal gate (can make any function)

Truth Table: XOR

- Input A | Input B | Output
- 0 | 0 | 0
- 0 | 1 | 1
- 1 | 0 | 1
- 1 | 1 | 0
- NAND truth table (2 inputs):
- $0,0 \rightarrow 1$ | $0,1 \rightarrow 1$
- $1,0 \rightarrow 1$ | $1,1 \rightarrow 0$



Part 2: Amplifier Classes

Subelement E7B — 1 exam question



Amplifier Classes Comparison

- Class A: device biased halfway between saturation and cutoff
 - → Conducts for the full 360° of the input cycle
 - → Lowest distortion; can amplify any waveform including SSB
 - → Lowest efficiency ($\sim 25\text{-}30\%$); wastes power as heat
- Class AB: each device in a push-pull pair conducts $>180^\circ$ but $<360^\circ$
 - → Good linearity; better efficiency than A ($\sim 50\text{-}70\%$)
 - → Standard for SSB linear amplifiers
- Class B: each device conducts exactly 180° (push-pull)
 - → $\sim 60\text{-}70\%$ efficiency; crossover distortion without bias
- Class C: conduction less than 180°
 - → Highest distortion — NOT suitable for SSB
 - → Works for CW and FM (constant envelope) with tuned output
- → Efficiency $>80\%$



Amplifier Configurations

- Common emitter: most common; voltage gain with phase inversion
 - → Figure E7-1: $R1/R2$ = voltage divider bias network; $R3$ = emitter self-bias
- Common collector (emitter follower):
 - → Input and output in PHASE (no phase inversion)
 - → Voltage gain ~ 1 ; high input impedance, low output impedance
 - → Used as impedance buffer
- Grounded-grid amplifier (vacuum tube):
 - → Low input impedance
 - → Used in high-power HF amplifiers
- Parasitic oscillation:
 - → Unintended oscillation in an amplifier stage
 - → Fix: install parasitic suppressors (ferrite beads on leads) and/or neutralize the stage



Part 3: Filters and Matching

Subelement E7C — 1 exam question



Filter Response Types

- Butterworth: maximally flat passband — no ripple anywhere
 - → Smooth rolloff; moderate transition steepness
 - → Best when passband flatness is most important
- Chebyshev: ripple in the passband
 - → Sharper cutoff than Butterworth for same number of poles
 - → Trades passband ripple for steeper transition
 - → Used when stopband attenuation is important and passband ripple is acceptable
- Elliptical (Cauer): ripple in BOTH passband and stopband
 - → Steepest transition of all for same number of poles
 - → Has transmission zeros (notches) in the stopband
 - → Used where sharp cutoff and stopband rejection are critical



Matching Networks and Stubs

- Pi-network (low-pass):
 - → Shunt C to ground → series L → shunt C to ground
 - → Standard amplifier output matching; provides harmonic suppression
- T-network (high-pass):
 - → Series C → shunt L to ground → series C
- Pi-L network: Pi-network + additional series inductor
 - → Greater harmonic suppression than Pi alone
- Band-pass filter: has two cutoff frequencies (upper and lower -3 dB points)
- Stub filter: short section of transmission line (open or shorted)
 - → Used at VHF/UHF/microwave for filtering



Part 4: Power Supplies and Modulation

Subelements E7D-E — 1 exam question



Modulation and Demodulation Circuits

Modulator Circuits

- Reactance modulator: varies oscillator frequency with audio → FM
- Phase modulator: varies phase → PM (phase modulation)
- Balanced modulator: multiplies carrier $\times$ audio
 - → Output: DSB with SUPPRESSED CARRIER
 - → Carrier is canceled; only sidebands remain
- SSB filter method:
 - → Balanced modulator + sideband filter = SSB
- Switching regulator: high efficiency; switches to transfer energy
- Linear regulator: simple but dissipates power as heat

Demodulator Circuits

- Product detector: multiplies SSB signal with BFO
 - → BFO reinserts the carrier; recovers audio
 - → Used to demodulate SSB and CW
- Mixer: multiplies two frequencies
 - → Output: sum (f_1+f_2) and difference (f_1-f_2)
 - → Used in superheterodyne receivers for frequency conversion
- Envelope detector: follows AM carrier amplitude
 - → Simple AM demodulator
- Solar array charge controller: manages voltage/current from solar panels to batteries



Part 5: SDR and DSP

Subelement E7F — 1 exam question



Software Defined Radio Architecture

- Traditional radio: analog filters and mixers do most of the work
- SDR: convert to digital as early as possible; do everything in software
- ADC (Analog-to-Digital Converter):
 - → Samples the analog signal at high rate
 - → Nyquist criterion: sample rate must be $\geq 2 \times$ highest signal frequency
 - → Quantization: rounds each sample to nearest digital step
 - → Dynamic range: ~ 6 dB per bit (16-bit = ~ 96 dB)
- SDR overload: input signal exceeds ADC reference voltage
 - → Fix: add attenuation before the ADC
- DSP filtering: performed AFTER analog-to-digital conversion (in software)
 - → FIR filter: finite impulse response; always stable; linear phase possible
 - → IIR filter: infinite impulse response; recursive; more efficient; can oscillate
- Advantages of DSP: flexible, reconfigurable, highly precise filter shapes



Part 6: Operational Amplifiers

Subelement E7G — 1 exam question



Op-Amp Configurations and Gain

- Ideal op-amp: infinite input impedance, zero output impedance, infinite open-loop gain
- Inverting amplifier:
 - $\rightarrow \text{Gain} = -(R_f/R_{in})$ [negative = phase inversion]
 - $\rightarrow$ Virtual ground at inverting input
 - $\rightarrow$ Example: $R_{in}=1\text{k}\Omega$, $R_f=10\text{k}\Omega \rightarrow \text{Gain} = -10$
- Non-inverting amplifier:
 - $\rightarrow \text{Gain} = 1 + (R_f/R_{in})$ [positive = in phase]
 - $\rightarrow$ Example: $R_{in}=1\text{k}\Omega$, $R_f=9\text{k}\Omega \rightarrow \text{Gain} = 1+9 = 10$
- Voltage follower (unity gain buffer):
 - $\rightarrow$ Output connected directly to inverting input



Part 7: Oscillators

Subelement E7H — 1 exam question



Oscillator Types

- Colpitts: capacitive voltage divider provides the feedback
 - → Two capacitors in the tank circuit; the junction between them feeds back
 - → Common RF oscillator; stable; widely used
- Hartley: inductive voltage divider (tapped coil) provides feedback
 - → Tapped inductor in the tank; tap point feeds back
 - → Simple; good for VFOs
- Crystal oscillator: quartz crystal controls frequency via mechanical resonance
 - → Very stable; used as reference oscillator
- VCO (Voltage Controlled Oscillator): varactor tunes the tank circuit
 - → Voltage changes capacitance; frequency follows
- PLL (Phase-Locked Loop): VCO output compared to reference; error corrects VCO
 - → Locks VCO to crystal-stable reference at any synthesized frequency
- DDS (Direct Digital Synthesizer): digital lookup table + DAC



Oscillator Quality and Stability

- TCXO (Temperature-Compensated Crystal Oscillator):
 - → Compensation circuit corrects for crystal frequency variation with temperature
 - → Used in GPS receivers, precision instruments
- OCXO (Oven-Controlled Crystal Oscillator):
 - → Crystal kept at constant elevated temperature in a small oven
 - → Highest practical frequency stability
- Phase noise:
 - → Close-in frequency instability — noise sidebands on either side of carrier
 - → Affects receiver performance: LO phase noise mixes with nearby strong signals
 - → Synthesizer phase noise degrades close-in signal quality
- Microphonics:
 - → Oscillator frequency varies with mechanical vibration
- → Classic problem in VFOs — physical shock changes the LC values slightly

Key Numbers — Module 5

Class A conduction	360° (full cycle); lowest distortion
Class C for SSB	NOT suitable — distorts the signal
Class D	Switching; >90% efficiency; needs output filter
Flip-flop divides by	2 (each stage); 4 stages = $\div 16$
Astable multivibrator	Continuously oscillates; no external clock
NAND output = 0 when	ALL inputs are 1
XOR output = 1 when	Inputs DIFFER
Butterworth filter	Flat passband; no ripple
Chebyshev filter	Ripple in passband; sharper cutoff
Elliptical filter	Ripple in both bands; steepest transition
Balanced modulator output	DSB suppressed carrier (no carrier)
Product detector function	Demodulates SSB
Nyquist rate	Sample $\geq 2\times$ highest signal frequency
ADC dynamic range	~ 6 dB per bit
Inverting op-amp gain	$-(R_f/R_{in})$
Non-inverting op-amp gain	$1 + (R_f/R_{in})$



Checkpoint Quiz 1 / 10

How many flip-flops are needed to divide a frequency by 16?

- A. 2
- B. 4
- C. 8
- D. 16

Answer: B — Each flip-flop divides by 2; $2^4=16$; need 4 stages



Checkpoint Quiz 2 / 10

Which amplifier class is NOT suitable for SSB?

- A. Class A
- B. Class AB
- C. Class C
- D. Class B

Answer: C — Class C conducts $<180^\circ$; distorts SSB (signal distortion and excessive bandwidth)



Checkpoint Quiz 3 / 10

What type of filter has ripple in both passband and stopband?

- A. Butterworth
- B. Bessel
- C. Chebyshev
- D. Elliptical

Answer: D — Elliptical (Cauer): ripple in BOTH passband and stopband; steepest transition



Checkpoint Quiz 4 / 10

What does a balanced modulator produce?

- A. AM with full carrier
- B. FM signal
- C. DSB with suppressed carrier
- D. SSB with pilot carrier

Answer: C — Balanced modulator cancels the carrier; produces Double Sideband Suppressed Carrier



Checkpoint Quiz 5 / 10

What is the dynamic range per bit in an ADC?

- A. 3 dB
- B. 6 dB
- C. 12 dB
- D. 20 dB

Answer: B — Approximately 6 dB per bit (more precisely 6.02 dB)



Checkpoint Quiz 6 / 10

What is the gain of an inverting op-amp with $R_{in}=2.2k\Omega$ and $R_f=22k\Omega$?

- A. +10
- B. -10
- C. +11
- D. -11

Answer: B — Gain = $-(R_f/R_{in}) = -(22k/2.2k) = -10$



Checkpoint Quiz 7 / 10

What is the gain of a non-inverting op-amp with $R_{in}=1k\Omega$ and $R_f=4.7k\Omega$?

- A. 4.7
- B. -4.7
- C. 5.7
- D. -5.7

Answer: C — $\text{Gain} = 1 + (R_f/R_{in}) = 1 + (4.7k/1k) = 5.7$



Checkpoint Quiz 8 / 10

A Colpitts oscillator uses which type of feedback network?

- A. Tapped inductor
- B. Capacitive voltage divider
- C. Crystal
- D. Phase-shift network

Answer: B — Colpitts: two capacitors form a voltage divider; the junction provides feedback



Checkpoint Quiz 9 / 10

What is the main advantage of a PLL synthesizer over a free-running VFO?

- A. Wider tuning range
- B. Simpler circuit
- C. Frequency stability referenced to a crystal
- D. Lower phase noise

Answer: C — PLL locks the VCO to a crystal reference; gives crystal stability across the tuning range



Checkpoint Quiz 10 / 10

An astable multivibrator is best described as:

- A. A circuit that toggles when triggered
- B. A circuit that continuously alternates between two states without external triggering
- C. A frequency divider
- D. A bistable memory element

Answer: B — Astable: no stable state; generates its own square wave continuously



Before Module 6

Your assignment:

- Drill HamStudy E7A-E7H to 80%+ confidence — this subelement takes the most time
- Practice the two op-amp gain formulas until they're automatic
- Know all three filter names and their passband/stopband ripple behavior

Coming up:

Module 6: Signals, Emissions & Safety — only 5 questions but includes formulas (Carson's rule, RMS voltage, modulation index). Good recovery module after this one.